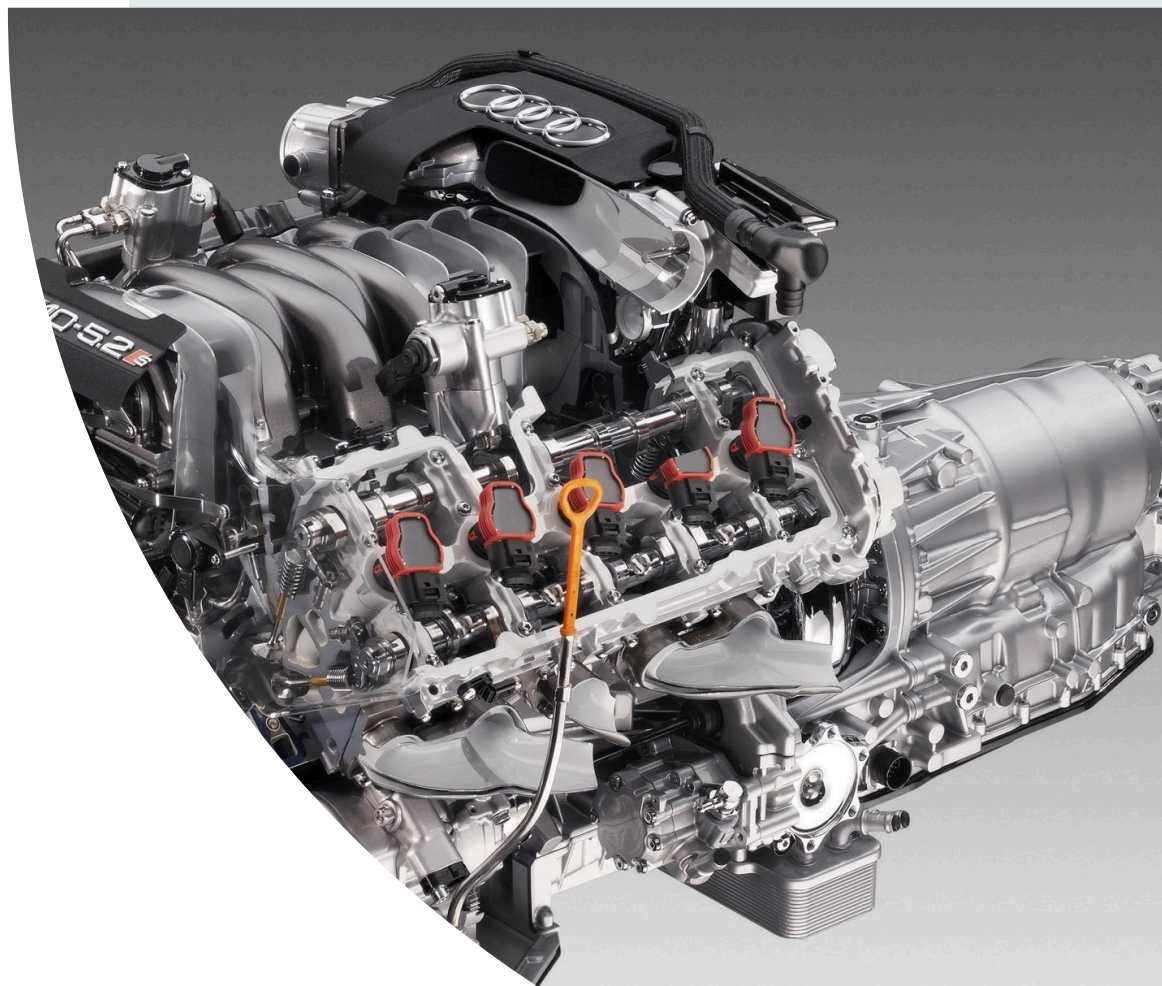


INTERNAL COMBUSTION ENGINES

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PRACTICAL REPORT ON INTERNAL
COMBUSTION ENGINES



INTRODUCTION

INTERNAL COMBUSTION ENGINES ARE ONE OF THE MOST PROMINENT INNOVATIONS IN THE FIELD OF MECHANICAL ENGINEERING, AS THEY REPRESENT THE HEART OF MANY INDUSTRIAL AND TRANSPORTATION APPLICATIONS. THESE ENGINES RELY ON BURNING FUEL INSIDE A CLOSED COMBUSTION CHAMBER TO PRODUCE MECHANICAL ENERGY, ALLOWING THE CONVERSION OF THE CHEMICAL ENERGY STORED IN THE FUEL INTO KINETIC ENERGY. INTERNAL COMBUSTION ENGINES ARE WIDELY USED IN CARS, AIRCRAFT, HEAVY EQUIPMENT, IN ADDITION TO .MARINE APPLICATIONS AND OTHERS

THE DESIGN OF THE ROTARY ENGINE AIMS TO PROVIDE AN EFFECTIVE ALTERNATIVE TO TRADITIONAL COMBUSTION ENGINES, WHICH OFTEN RELY ON THE RECIPROCATING MOTION OF PISTONS. THE ROTARY ENGINE IS CHARACTERIZED BY ITS INNOVATIVE DESIGN THAT ALLOWS FOR HIGHER PERFORMANCE AND BETTER EFFICIENCY, WHILE REDUCING THE RESULTING VIBRATIONS AND NOISE. BY USING A ROTARY DESIGN, THE DYNAMICS OF THE ENGINE CAN BE IMPROVED, WHICH LEADS TO IMPROVED FUEL CONSUMPTION AND INCREASED OUTPUT POWER. THIS DESIGN AIMS TO MEET THE GROWING MARKET DEMANDS FOR CLEAN AND EFFICIENT ENERGY, WHICH CONTRIBUTES TO DRIVING INNOVATION IN THE FIELD OF .ENGINE TECHNOLOGY



DESIGN DESCRIPTION

1. WORKING PRINCIPLE

THE ROTARY ENGINE WORKS ON THE PRINCIPLE OF FUEL COMBUSTION INSIDE A CLOSED CHAMBER, WHERE A ROTARY DESIGN IS USED INSTEAD OF TRADITIONAL PISTONS. THE ENGINE USUALLY CONSISTS OF A THREE-PHASE ROTOR SIMILAR TO A FAN AND ROTATES AROUND A FIXED AXIS, WHICH LEADS TO REDUCING RECIPROCATING MOTION AND REDUCING .VIBRATIONS

BASIC COMPONENTS .2

EXTERNAL STRUCTURE: FORMS THE BASE OF THE ENGINE AND IS MADE OF STRONG MATERIALS SUCH AS CAST IRON OR ALUMINUM TO .WITHSTAND THE PRESSURE OF COMBUSTION

ROTOR: THE ROTATING PART OF THE ENGINE, WHICH CONSISTS OF A THREE-DIMENSIONAL SHAPE OR MORE, AS IT CONTAINS THREE SIDES .THAT FORM DIFFERENT COMBUSTION CHAMBERS IN EACH CYCLE

COMBUSTION CHAMBER: CONTAINS FUEL AND AIR, WHERE THE FUEL IS BURNED TO PRODUCE ENERGY. IT IS DESIGNED TO BE OF A SUITABLE SIZE .TO ACHIEVE OPTIMAL COMBUSTION EFFICIENCY

IGNITION SYSTEM: CONSISTS OF SPARK PLUGS OR AN ELECTRONIC IGNITION SYSTEM THAT IGNITES THE AIR-FUEL MIXTURE AT THE RIGHT .TIME

LUBRICATION SYSTEM: USED TO ENSURE THAT MOVING PARTS OPERATE SMOOTHLY AND REDUCE FRICTION. IT CAN BE A DIRECT LUBRICATION .SYSTEM OR BASED ON AN OIL CYCLE

OUTLETS: THESE INCLUDE THE EXHAUST OUTLET, WHICH RELEASES THE COMBUSTION GASES, AND THE POWER OUTLET, WHICH TRANSFERS THE RESULTING ENERGY TO OTHER COMPONENTS, SUCH AS THE WHEELS IN .THE VEHICLE

3. STRUCTURAL DESIGN

REINFORCED BODY: DESIGNED TO BE LIGHTWEIGHT AND STRONG, ADVANCED MANUFACTURING TECHNIQUES ARE USED TO IMPROVE THE .STRENGTH OF THE BODY WHILE REDUCING WEIGHT

DYNAMIC BALANCE: THE ROTATING DESIGN HELPS ACHIEVE BETTER .DYNAMIC BALANCE, REDUCING VIBRATIONS CAUSED BY MOVEMENT

MATERIALS USED .4

MATERIALS ARE CAREFULLY SELECTED TO WITHSTAND HIGH TEMPERATURES AND COMBUSTION PRESSURES, SUCH AS STAINLESS STEEL FOR THE INTERIOR PARTS, AND ALUMINUM OR LIGHT ALLOYS FOR .THE REST OF THE BODY

POLLUTION CONTROL .5

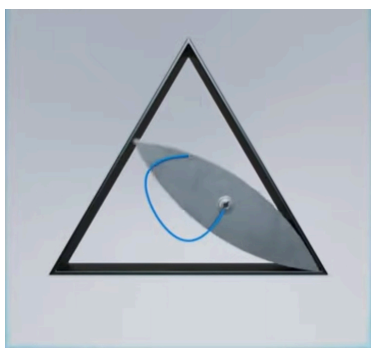
EXHAUST EMISSION CONTROL SYSTEMS, SUCH AS CATALYTIC CONVERTERS, CAN BE INTEGRATED TO ENSURE COMPLIANCE WITH .ENVIRONMENTAL STANDARDS

CONCLUSION

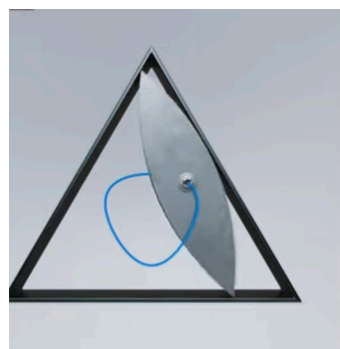
THE ENGINE DESIGN IS INNOVATIVE AND EFFICIENT, MAKING IT A POPULAR CHOICE IN MULTIPLE AREAS THAT REQUIRE STRONG .PERFORMANCE WHILE REDUCING NOISE AND VIBRATION



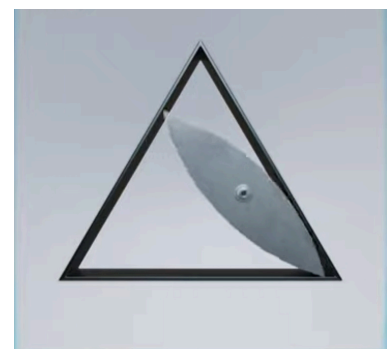
INTAKE



COMPRESSION



POWER



EXHAUST

PERFORMANCE AND EFFICIENCY

1. THERMAL EFFICIENCY

COMPRESSION RATIO: ROTARY ENGINES TYPICALLY HAVE HIGHER COMPRESSION RATIOS THAN CONVENTIONAL COMBUSTION ENGINES. THIS HELPS IMPROVE THERMAL EFFICIENCY, AS MORE OF THE ENERGY GENERATED BY THE COMBUSTION OF FUEL IS UTILIZED

IMPROVED COMBUSTION: THE DESIGN OF THE COMBUSTION CHAMBER HELPS ACHIEVE MORE EFFICIENT COMBUSTION, REDUCING FUEL CONSUMPTION AND INCREASING POWER OUTPUT

POWER OUTPUT .2

HIGH POWER IN A SMALL VOLUME: ROTARY ENGINES ARE CAPABLE OF DELIVERING MORE POWER COMPARED TO THEIR SIZE. OUTSTANDING PERFORMANCE CAN BE ACHIEVED WITH REDUCED WEIGHT, MAKING THEM IDEAL FOR USE IN SPORTS CARS AND LIGHT AIRCRAFT

QUICK RESPONSE: ROTARY ENGINES PROVIDE QUICK THROTTLE RESPONSE, PROVIDING A SENSE OF IMMEDIATE CONTROL AND SMOOTH ACCELERATION

FRICTION AND VIBRATION .3

REDUCED FRICTION: DUE TO THEIR ROTARY DESIGN, FRICTION FROM MOVING PARTS IS REDUCED, IMPROVING OVERALL EFFICIENCY AND REDUCING POWER LOSS

LESS VIBRATION: ROTARY ENGINES PRODUCE LESS VIBRATION COMPARED TO CONVENTIONAL RECIPROCATING ENGINES, RESULTING IN A SMOOTHER AND MORE COMFORTABLE DRIVING EXPERIENCE

4. EXHAUST EMISSIONS

IMPROVED EMISSION CONTROL: THE TECHNOLOGY USED IN THE DESIGN OF THE COMBUSTION CHAMBER AND IGNITION SYSTEM CAN HELP REDUCE HARMFUL EXHAUST EMISSIONS, SUCH AS NITROGEN OXIDES (NOX) AND .UNBURNED HYDROCARBONS

ENVIRONMENTAL COMPLIANCE: ALTHOUGH IT MAY REQUIRE SOME MODIFICATIONS, THE ROTARY ENGINE CAN BE DESIGNED TO COMPLY WITH MODERN .ENVIRONMENTAL STANDARDS

FUEL EFFICIENCY .5

LOWER FUEL CONSUMPTION: THE ROTARY ENGINE TENDS TO BE MORE FUEL EFFICIENT UNDER NORMAL CONDITIONS THAN SOME CONVENTIONAL COMBUSTION ENGINES, MAKING IT A PREFERRED CHOICE IN APPLICATIONS THAT .REQUIRE ENERGY EFFICIENCY



TORQUE (T): 150 LB-FT.

ROTATION SPEED 4000 RPM (RPM)

ENGINE SIZE (V): 1.5 LITERS (I.E. 0.0015 CUBIC METERS)

.COMPRESSION RATIO (CR) 10:1

COMBUSTION PRESSURE (150 PSI)

.ENERGY OUTPUT FROM COMBUSTION OF 1 LITER OF FUEL 34

THERMAL EFFICIENCY 0.3

CALCULATING THE OUTPUT POWER (P) .1

:WE USE THE EQUATION

$$P = 5252$$

:FIRST, WE CALCULATE THE OUTPUT POWER

$$P = 150 \times 4000 \times 5252 \approx 114.2$$

CALCULATING THE COMPRESSION RATIO 2

:THE COMPRESSION RATIO (CR) IS 10:1, WHICH MEANS

$$CR = V_{MAX} / V_{MIN} = 10$$

3 CALCULATING FUEL CONSUMPTION

**FIRST, WE CALCULATE THE ENERGY PRODUCED BY
BURNING 1.5 LITERS OF**

:FUEL

**$Q = 34 \text{ MJ} \times 1.5 \approx 51 \text{ MJ} = 51,000,000 \text{ J}$ NOW WE CALCULATE
THE FUEL CONSUMPTION**

$170,000,000 \approx 51,000,000/3.5$

CALCULATING THE POWER GENERATED BY THE ENGINE 4

:WE USE THE EQUATION

$$I = VP^2$$

SINCE 150 = MUST BE CONVERTED TO THE UNIT

SI):

$P 150 \text{ PSI} \times 6894.76 \text{ PA/PSI} \approx 1,033,140 \text{ PA}$

NOW WE CALCULATE THE POWER GENERATED

$N 776.4 \approx 2 / 1,033,140 \times 0.0015 =$

:SUMMARY OF RESULTS

.OUTPUT POWER: 114.2 HP

.COMPRESSION RATIO: 10:1

FUEL CONSUMPTION: 170,000,000 J •

OUTPUT POWER: 776.4 N

CHALLENGES AND SOLUTIONS

:CHALLENGES

HIGH TEMPERATURES: THE ENGINE OPERATES UNDER EXTREME THERMAL .CONDITIONS, WHICH CAN LEAD TO WEAR AND TEAR ON PARTS

HIGH FUEL CONSUMPTION: DESPITE EFFICIENCY, FUEL CONSUMPTION .CAN BE HIGH IN SOME CASES

EMISSION CONTROL: COMPLIANCE WITH ENVIRONMENTAL STANDARDS .REQUIRES EFFECTIVE SYSTEMS TO CONTROL HARMFUL EMISSIONS

DESIGN COMPLEXITY: ROTARY DESIGN REQUIRES HIGH PRECISION IN .MANUFACTURING TO ENSURE OPTIMAL PERFORMANCE

:SOLUTIONS

ADVANCED COOLING SYSTEMS: USE EFFICIENT COOLING SYSTEMS TO .REDUCE TEMPERATURES INSIDE THE ENGINE

IMPROVING COMBUSTION DESIGN: DEVELOPING THE COMBUSTION CHAMBER DESIGN TO IMPROVE FUEL COMBUSTION EFFICIENCY AND .REDUCE CONSUMPTION

IMPLEMENTING ADVANCED TECHNOLOGIES: USING TECHNOLOGIES SUCH .AS CATALYTIC CONVERTERS TO REDUCE EMISSIONS

RESEARCH AND DEVELOPMENT: INVESTING IN RESEARCH TO IMPROVE .MANUFACTURING TECHNOLOGIES AND REDUCE COMPLEXITY

POTENTIAL APPLICATIONS OF THE ENGINE

**AUTOMOTIVE: IT IS USED IN SPORTS CARS AND SMALL CARS DUE TO ITS
.SMALL SIZE AND LIGHT WEIGHT**

**LIGHT AIRCRAFT: IT PROVIDES POWERFUL POWER IN A COMPACT SIZE,
.MAKING IT SUITABLE FOR LIGHT AIRCRAFT**

**MARINE EQUIPMENT: IT CAN BE USED IN BOATS AND SHIPS THAT REQUIRE
.EFFICIENT AND HIGH-PERFORMANCE ENGINES**

**POWER GENERATION: IT CAN BE USED IN INDUSTRIAL APPLICATIONS TO
.GENERATE ELECTRICAL ENERGY**



CONCLUSION

THE ENGINE IS A PRIME EXAMPLE OF INNOVATION IN ENGINE TECHNOLOGY, COMBINING HIGH PERFORMANCE AND OUTSTANDING EFFICIENCY IN A COMPACT DESIGN. DESPITE THE CHALLENGES IT FACES, THE AVAILABLE SOLUTIONS AND ITS WIDE APPLICATIONS MAKE IT AN ATTRACTIVE OPTION IN VARIOUS FIELDS. WITH CONTINUED RESEARCH AND DEVELOPMENT, THE ROTARY ENGINE CAN ACHIEVE FURTHER SUCCESS IN MEETING INCREASING POWER DEMANDS WHILE MAINTAINING COMPLIANCE WITH .ENVIRONMENTAL STANDARDS